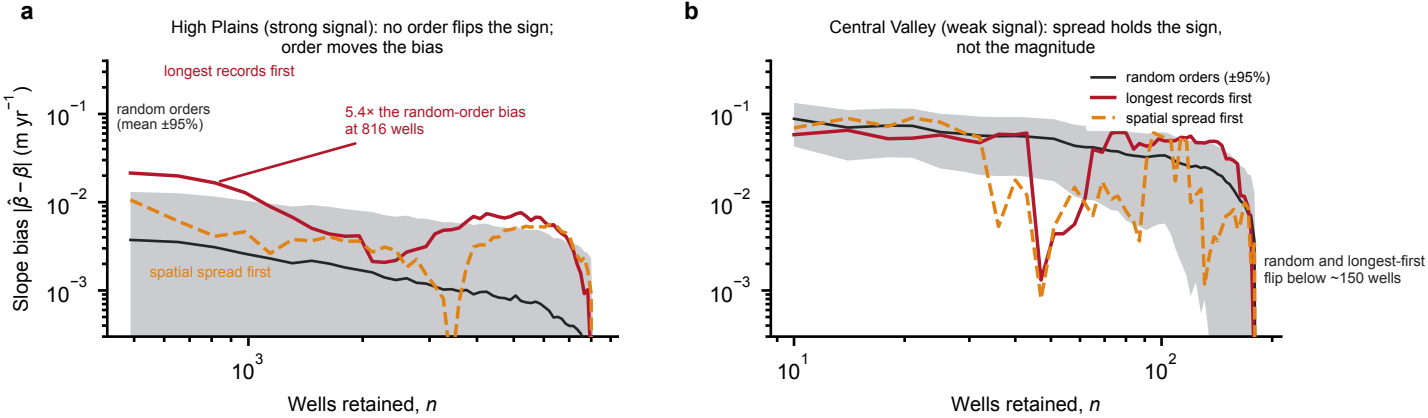
a
High Plains (strong signal): no order flips the sign;
order moves the bias
longest records first
random orders
(mean ±95%)
5.4× the random-order bias
at 816 wells
spatial spread first
Slope bias $|\hat{\beta} - \beta|$ (m yr^{-1})
10^{-1}
10^{-2}
10^{-3}
10^{3}
Wells retained, n
b
Central Valley (weak signal): spread holds the sign,
not the magnitude
random orders (±95%)
longest records first
spatial spread first
random and longest-first
flip below ~150 wells
10^{-1}
10^{-2}
10^{-3}
10^{1}
10^{2}
Wells retained, n